

SOAR Platform Technical Architecture

Security Orchestration, Automation, and Response

1. Executive Summary

The SOAR (Security Orchestration, Automation, and Response) platform is an advanced AI-powered security automation system designed to augment existing SIEM solutions by intelligently processing security alerts, reducing false positives through machine learning analysis, and automating incident response actions.

1.1 System Overview

The platform processes an average of 15,821 alerts per day (with 30% growth margin) from multiple SIEM sources including Wazuh, SentinelOne, Sumo Logic, and Splunk. It leverages 8 machine learning models and 5 AI agents to provide intelligent, context-aware security automation.

1.2 Key Capabilities

Capability	Description
Cross-Device Correlation	Groups related alerts across multiple hosts into attack chains
False Positive Detection	ML-powered identification and filtering of false alarms
Behavioral Anomaly Detection	Identifies unusual patterns in alerts, users, and assets
Predictive Attack Forecasting	Predicts attack windows and next stages in ongoing attacks
Automated Root Cause Analysis	Identifies attack entry points and root causes automatically
Dynamic Response Optimization	Learns optimal response actions through reinforcement learning

1.3 Performance Metrics

Metric	Value
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Alert Processing Capacity	50,000 alerts/day (3.2x current load)
Average Processing Latency	3-5 seconds per alert (end-to-end)
False Positive Detection Rate	>85% accuracy (target)
System Uptime	99.5% (target)

2. System Architecture

2.1 High-Level Architecture

The SOAR platform follows a microservices architecture with 10 distinct processing layers:

Layer 1: SIEM Integration & Ingestion

- Receives alerts from Wazuh, SentinelOne, Sumo Logic, and Splunk via REST API webhooks.
- Load balancer (NGINX) distributes traffic across multiple FastAPI ingestion service instances.

Layer 2: Message Queue (Redis Streams)

- Queues: alerts:incoming (main), alerts:dlq (dead letter), alerts:priority (high-priority bypass).
- Ensures message ordering and implements backpressure handling.

Layer 3: Context Enrichment (Parallel)

- Three parallel services: (1) Threat Intel (VirusTotal, OTX, AbuseIPDB), (2) GeoIP (MaxMind), (3) Asset Context (criticality, compliance scope).
- Results cached in Redis.

Layer 4: ML Model Processing (Sequential)

- Sequential execution: Model 1 (Anomaly Detection) → Model 2 (FP Classification) → Model 3 (Root Cause Analysis).
- Model 4 (Attack Chain Detection) runs as background service every 5 minutes.

Layer 5: RAG Document Retrieval

- Vector database (Qdrant) retrieves top-5 relevant documents from MITRE ATT&CK, security playbooks, and historical incidents using semantic similarity search.

Layer 6: AI Agent Analysis

- Agent 1 (Alert Analysis) combines ML scores, enrichment, and RAG docs, then calls LLM (Model 8) for reasoning.

- Agent 3 (Verification) validates outputs. Agent 4 (RL Optimizer) recommends optimal action.

Layer 7: Decision & Action Engine

- Fuses recommendations from all agents, applies policy gates, and executes approved actions via Response Executor (firewall API, EDR API, ticketing system, notifications).

Layer 8: Data Storage

- MongoDB stores alerts_raw, alerts_processed, actions_taken, model_predictions.
- Redis caches frequently accessed data. Qdrant maintains vector embeddings.

Layer 9: Dashboard & Visualization

- React dashboard with WebSocket for real-time updates.
- REST API for historical queries. Supports manual review, analytics, and reporting.

Layer 10: Feedback & Learning Loop

- Collects analyst corrections and action outcomes.
- Feeds back to weekly ML model retraining. RL agent updates continuously from real-world results.

2.2 Processing Flow Diagram

Main Alert Processing Path (Synchronous - 3-5 seconds):

Wazuh SIEM → Load Balancer → Ingestion Service (6 functions) → Redis Queue (3 functions) → Enrichment Services [Threat Intel + GeoIP + Asset Context] (12 functions) → ML Models [Anomaly + FP + Root Cause] (15 functions) → RAG Retrieval (4 functions) → Agent 1: LLM Analysis (5 functions) → Agent 3: Verification (6 functions) → Agent 4: RL Optimizer (3 functions) → Decision Engine (4 functions) → Response Executor (2 functions) → Storage (6 functions) → Dashboard Update (4 functions) → Feedback Loop (async)

Background Processes (Asynchronous):

- Model 4 (Attack Chain Detection): Runs every 5 minutes
- Agent 2 (Threat Hunting): Runs every hour
- Agent 5 (Predictive Forecasting): Runs every 6 hours
- Model Retraining: Weekly (or on-demand)
- Feedback Collection: Continuous

3. Machine Learning Models

The platform employs 8 specialized ML models for different aspects of threat detection and response:

Model	Algorithm	Purpose	
1. Anomaly Detector	Isolation Forest	Detect unusual patterns	
2. FP Classifier	XGBoost	Identify false positives	
3. Root Cause	Random Forest	Identify attack origin	
4. Attack Chain	Graph Clustering	Correlate multi-host attacks	
5. Time Forecaster	Prophet	Predict attack windows	
6. Stage Predictor	LSTM	Predict next attack stage	
7. Response Optimizer	PPO (RL)	Learn optimal actions	
8. LLM	Llama-3-8B	Reasoning & analysis	

** Model 2 (FP Classifier) requires 500+ labeled alerts collected during Weeks 1-8*

3.1 Model Details

Model 1: Anomaly Detector (Isolation Forest)

Attribute	Detail
Algorithm	Isolation Forest (sklearn)
Features	48 features (temporal, rule-based, network, user/asset, contextual, behavioral)

Training Data	30 days normal alerts (10K-50K samples, unlabeled)
Output	Anomaly score (0-1), binary classification, confidence
Latency	<10ms per alert
Deployment	Week 4 - anomaly-detector:5001

Model 8: Large Language Model (Llama-3-8B-Instruct)

Attribute	Detail
Model	Llama-3-8B-Instruct (Quantized Q4_K_M)
Context Window	4096 tokens
GPU Required	NVIDIA RTX 4090 (24GB) - can run 2 instances simultaneously
Throughput	~30 alerts/minute per instance
Latency	<3 seconds per alert
Purpose	Powers Agent 1 (Alert Analysis) and Agent 2 (Threat Hunting)

4. AI Agents

The platform deploys 5 intelligent AI agents that leverage the ML models for automated decision-making:

Agent	Models Used	Purpose

1. Alert Analysis	1, 2, 3, 8	Primary analysis - classifies alerts and recommends actions
2. Threat Hunting	1, 8	Proactive threat discovery through autonomous hunting
3. Verification	1, 2	Quality control - validates Agent 1 outputs, prevents hallucinations
4. RL Optimizer	7	Learns optimal response actions from outcomes
5. Forecasting	5, 6	Predicts attack windows and next attack stages

4.1 Agent 1: Alert Analysis Agent (Primary)

The core intelligence agent that processes every incoming alert through a multi-step analysis pipeline:

Processing Steps:

Step 1: Context Gathering

- Retrieves enrichment data (threat intel, GeoIP, asset context)
- Collects ML scores from Models 1, 2, 3
- Retrieves relevant documents from vector DB
- Gathers user behavioral analytics signals

Step 2: LLM Analysis

- Builds comprehensive prompt combining all context
- Calls Model 8 (LLM) with structured prompt
- Receives JSON output with classification, severity, recommended action
- Includes detailed justification and reasoning

Step 3: Output Formatting

- Combines LLM response with ML scores
 - Calculates overall confidence score
 - Adds metadata (alert_id, timestamp, processing time)
 - Passes to Agent 3 for verification
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5. Infrastructure Requirements

5.1 Production Hardware Specifications

Based on 15,821 alerts/day with 30% growth margin:

Server 1: Processing Server

Component	Specification
CPU	AMD EPYC 7543P (32 cores, 2.8 GHz)
RAM	128 GB DDR4-3200 ECC
Storage	1 TB NVMe + 2 TB SSD
GPU	NVIDIA RTX 4090 (24 GB)
Network	10 Gbps NIC

Server 2: Data Server

Component	Specification
CPU	AMD EPYC 7313P (20 cores, 3.0 GHz)
RAM	128 GB DDR4-3200 ECC
Storage	500 GB NVMe + 2 TB NVMe + 4 TB SSD + 8 TB HDD
Network	10 Gbps NIC
RAID	RAID 10 (SSD), RAID 5 (HDD)

5.2 Storage Requirements

Component	Retention
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Alert Storage (MongoDB)	90 days
Raw Logs (Elasticsearch)	90 days
Network Flows	30 days
Endpoint Telemetry	30 days
ML Models	Permanent
Vector DB (Qdrant)	Permanent
Redis Cache	Volatile

6. Deployment Architecture

6.1 Service Distribution

Server 1 (Processing Server) Services:

- Ingestion Service (3 instances, port 8001-8003)
- ML Model Servers (8 services, ports 5001-5008)
- LLM Service (2 instances, port 5006)
- Decision Engine (port 8010)
- Response Executor (port 8011)
- Redis Cache (if not on Server 2)

Server 2 (Data Server) Services:

- MongoDB (port 27017)
- Elasticsearch (port 9200)
- Redis (port 6379)
- Qdrant Vector DB (port 6333)
- Dashboard/WebSocket Server (port 3000/3001)
- Backup Services

6.2 Network Architecture

Layer	Specification	Bandwidth
Internal Network	Server 1 ↔ Server 2	10 Gbps
External Inbound	Wazuh → SOAR	1 Gbps (sufficient)
External Outbound	SOAR → APIs/Firewall/Dashboard	1 Gbps (sufficient)

7. Security & Compliance

7.1 Security Controls

Authentication & Authorization:

- API Key validation for all incoming requests
- HMAC signature verification for webhooks
- IP whitelisting for SIEM sources
- Role-based access control (RBAC) for dashboard users
- JWT tokens for API authentication

Data Protection:

- Encryption at rest (MongoDB, Elasticsearch)
- TLS 1.3 for all network communication
- Encrypted backups with AES-256
- Secure credential storage (HashiCorp Vault)
- Log data sanitization (PII removal)

Audit & Compliance:

- All actions logged to actions_taken collection
 - Immutable audit trail with timestamps
 - 90-day retention for compliance (configurable)
 - SOC 2 Type II alignment
 - GDPR data protection mechanisms
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8. Operational Procedures

8.1 Monitoring & Observability

Metrics Collection (Prometheus):

- Alert processing latency (p50, p95, p99)
- ML model inference time
- Queue depth and backpressure
- Database query performance
- API endpoint response times
- GPU utilization and memory
- System resource usage (CPU, RAM, disk)

Dashboards (Grafana):

- Real-time alert processing metrics
- ML model performance dashboard
- System health overview
- Capacity planning dashboard
- SLA compliance tracking

8.2 Backup & Recovery

Component	Frequency	Retention
MongoDB	Daily (incremental), Weekly (full)	30 days
ML Models	After each training	90 days
Configuration	On change	Indefinite

9. Development & Deployment Timeline

12-Week MVP Timeline (January 8 - March 31, 2026)

Phase	Weeks	Deliverables
Phase 1: Foundation	1-3	Infrastructure, ingestion, queue, feedback UI

Phase 2: Core ML	4	Models 1 & 5 (Anomaly, Prophet)
Phase 3: Behavioral	5	Models 3 & 4 (Root Cause, Attack Chain)
Phase 4: Advanced	6	Models 6, 7, 8 (LSTM, RL, LLM) + All 5 Agents
Phase 5: Integration	7-8	End-to-end testing, optimization
Phase 6: Training	9	Train Model 2 with 500+ labels, retrain others
Phase 7: Launch	10-12	UAT, production deployment, monitoring

Appendix: Glossary

Term	Definition
SOAR	Security Orchestration, Automation, and Response
SIEM	Security Information and Event Management
ULF	Unified Log Format - Normalized alert schema
ML	Machine Learning
LLM	Large Language Model
RL	Reinforcement Learning
RAG	Retrieval-Augmented Generation
MITRE ATT&CK	Framework of adversary tactics and techniques
FP	False Positive - Alert that is not a real threat
EDR	Endpoint Detection and Response